

Dedicated built-in-SM $d\bar{d} \rightarrow Z + \text{gluon}$ performance

Apple M3 Pro investigation

pyAmpliCol performance study F

Current $n = 8-9$ source `eee440107f3ddb4fa0fbe8e55dd3ed9702ee0e36`

This final investigation-F table retains the existing $n = 1-7$ evidence and reports every requested $n = 8-9$ terminal state. Successful cells are shown as measurements; capped cells show the independently reached resource limit. Catalog-declared ASM O3 and C++ O3 rows above $n = 6$ remain static N/A without an attempt or compiler launch.

Every new $n = 8-9$ attempt is guarded at exactly 30,000,000,000 bytes by the maximum of aggregate process-tree RSS and Darwin physical footprint, with its generation phase capped at 3600 seconds. Selected-flow rows use topology-replay with a helicity sum; all-flow rows use all-flow-union with one deterministic helicity. Runtime values are validated at the canonical deterministic phase-space point before timing. Every displayed pyAmpliCol mode reports its evaluator-total timing where that evidence exists.

6.1 Built-in SM dedicated $d\bar{d} \rightarrow Z + \text{gluon}$ performance

Built-in SM $d\bar{d} \rightarrow Z + \text{gluon}$ ladder (block 1 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [us/pt]	eval total [us/pt]	gen [s]	wall [us/pt]	eval total [us/pt]
1 AmpliCol reference	2.18	0.0217	NOT EXPOSED	5.3e-05	0.0475	NOT EXPOSED
1 compiled JIT O1	2 (x0.919)	0.181 (x8.33)	T 0.181	C 2; N.C.	0.086 (x1.81)	T 0.086
1 compiled ASM O3	4.71 (x2.16)	0.178 (x8.21)	T 0.178	C 3.83; N.C.	0.0852 (x1.8)	T 0.0852
1 compiled C++ O3	4.59 (x2.11)	0.181 (x8.35)	T 0.181	C 3.66; N.C.	0.083 (x1.75)	T 0.083
1 compiled JIT O3	1.99 (x0.911)	0.174 (x8.04)	T 0.174	C 1.96; N.C.	0.0847 (x1.78)	T 0.0847
1 eager-DAG JIT O2	2.26 (x1.04)	0.288 (x13.3)	T 0.288	C 2.2; N.C.	0.181 (x3.81)	T 0.181
1 recurrence JIT O2	3.97 (x1.82)	0.168 (x7.74)	T 0.168	C 3.81; N.C.	0.108 (x2.28)	T 0.108
2 AmpliCol reference	2.18	0.126	NOT EXPOSED	0.000104	0.115	NOT EXPOSED
2 compiled JIT O1	2.2 (x1.01)	0.39 (x3.09)	T 0.39	C 1.91; N.C.	0.145 (x1.26)	T 0.145
2 compiled ASM O3	6.44 (x2.95)	0.418 (x3.32)	T 0.418	C 4.53; N.C.	0.129 (x1.12)	T 0.129
2 compiled C++ O3	6.77 (x3.1)	0.427 (x3.39)	T 0.427	C 4.46; N.C.	0.13 (x1.13)	T 0.13
2 compiled JIT O3	2.26 (x1.04)	0.363 (x2.88)	T 0.363	C 2.01; N.C.	0.137 (x1.19)	T 0.137
2 eager-DAG JIT O2	2.26 (x1.04)	0.685 (x5.43)	T 0.685	C 2.16; N.C.	0.294 (x2.55)	T 0.294
2 recurrence JIT O2	3.77 (x1.73)	0.446 (x3.54)	T 0.446	C 3.71; N.C.	0.283 (x2.46)	T 0.283
3 AmpliCol reference	2.08	0.447	NOT EXPOSED	0.000294	0.312	NOT EXPOSED
3 compiled JIT O1	2.46 (x1.18)	0.703 (x1.57)	T 0.703	C 2.11; N.C.	0.281 (x0.901)	T 0.281
3 compiled ASM O3	13.8 (x6.65)	1.1 (x2.47)	T 1.1	C 8.11; N.C.	0.258 (x0.828)	T 0.258
3 compiled C++ O3	13.7 (x6.62)	1.08 (x2.41)	T 1.08	C 8.1; N.C.	0.26 (x0.835)	T 0.26
3 compiled JIT O3	2.46 (x1.19)	0.646 (x1.44)	T 0.646	C 2.11; N.C.	0.258 (x0.829)	T 0.258
3 eager-DAG JIT O2	2.15 (x1.04)	1.41 (x3.14)	T 1.41	C 2.8; N.C.	0.629 (x2.02)	T 0.629
3 recurrence JIT O2	3.75 (x1.81)	1.18 (x2.63)	T 1.18	C 3.73; N.C.	0.887 (x2.85)	T 0.887

Original AmpliCol is the denominator. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value marked n.c.; n.c. means not comparable because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no separately reported evaluator-total timing. Every pyAmpliCol row shows the authenticated accumulated warmed evaluator total marked T; T is not an attribution ratio. Recurrence core/execution attribution remains a separate metric in raw evidence and is not relabeled as evaluator total. Older entries without authenticated total evidence remain marked not exposed; it is not a missing measurement. user cap: native C++/ASM generation is not attempted above n=6 (native-backend-generation-cap-n6-v1).

Built-in SM $d\bar{d} \rightarrow Z + \text{gluon ladder}$ (block 2 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [us/pt]	eval total [us/pt]	gen [s]	wall [us/pt]	eval total [us/pt]
4 AmpliCol reference	2.16	1.41	NOT EXPOSED	0.00148	1.15	NOT EXPOSED
4 compiled JIT O1	3.8 (x1.76)	1.91 (x1.36)	T 1.91	C 2.79; N.C.	0.739 (x0.645)	T 0.739
4 compiled ASM O3	36 (x16.7)	4.39 (x3.11)	T 4.39	C 22.3; N.C.	1.15 (x1.01)	T 1.15
4 compiled C++ O3	35.8 (x16.6)	4.43 (x3.14)	T 4.43	C 21.6; N.C.	1.15 (x1)	T 1.15
4 compiled JIT O3	3.82 (x1.77)	1.76 (x1.25)	T 1.76	C 2.82; N.C.	0.692 (x0.604)	T 0.692
4 eager-DAG JIT O2	2.22 (x1.03)	4.1 (x2.91)	T 4.1	C 2.51; N.C.	1.95 (x1.7)	T 1.95
4 recurrence JIT O2	3.62 (x1.67)	2.87 (x2.04)	T 2.87	C 3.78; N.C.	4.35 (x3.8)	T 4.35
5 AmpliCol reference	2.46	4.66	NOT EXPOSED	0.012	6.62	NOT EXPOSED
5 compiled JIT O1	10.8 (x4.39)	7.86 (x1.69)	T 7.86	C 8.23; N.C.	3.62 (x0.546)	T 3.62
5 compiled ASM O3	141 (x57.1)	17.7 (x3.8)	T 17.7	C 111; N.C.	14.4 (x2.17)	T 14.4
5 compiled C++ O3	147 (x59.7)	22.5 (x4.82)	T 22.5	C 109; N.C.	13.4 (x2.02)	T 13.4
5 compiled JIT O3	11.6 (x4.7)	5.62 (x1.21)	T 5.62	C 8.1; N.C.	3.33 (x0.502)	T 3.33
5 eager-DAG JIT O2	2.56 (x1.04)	12 (x2.57)	T 12	C 5.04; N.C.	11.2 (x1.69)	T 11.2
5 recurrence JIT O2	4.07 (x1.65)	6.08 (x1.3)	T 6.08	C 4.28; N.C.	8.3 (x1.25)	T 8.3
6 AmpliCol reference	3.81	14.7	NOT EXPOSED	0.113	42	NOT EXPOSED
6 compiled JIT O1	66 (x17.3)	16.5 (x1.12)	T 16.5	C 57.3; N.C.	23.6 (x0.561)	T 23.6
6 compiled ASM O3	883 (x231)	73.8 (x5.02)	T 73.8	C 647; N.C.	174 (x4.13)	T 174
6 compiled C++ O3	807 (x212)	71.8 (x4.89)	T 71.8	C 652; N.C.	171 (x4.07)	T 171
6 compiled JIT O3	67.4 (x17.7)	15.9 (x1.08)	T 15.9	C 64.2; N.C.	22.1 (x0.527)	T 22.1
6 eager-DAG JIT O2	2.67 (x0.699)	31.3 (x2.13)	T 31.3	C 55.4; N.C.	81 (x1.93)	T 81
6 recurrence JIT O2	4.16 (x1.09)	14.9 (x1.01)	T 14.9	C 5.43; N.C.	43.4 (x1.03)	T 43.4

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Built-in SM $d\bar{d} \rightarrow Z + \text{gluon ladder}$ (block 3 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [us/pt]	eval total [us/pt]	gen [s]	wall [us/pt]	eval total [us/pt]
7 AmpliCol reference	5.08	37.8	NOT EXPOSED	1.23	315	NOT EXPOSED
7 compiled JIT O1	17.5 (x3.44)	49.4 (x1.31)	T 49.4	C 646; N.C.	258 (x0.819)	T 258
7 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled JIT O3	16.6 (x3.27)	46 (x1.22)	T 46	C 657; N.C.	232 (x0.735)	T 232
7 eager-DAG JIT O2	3.35 (x0.659)	83.4 (x2.21)	T 83.4	C 2.96e+03; N.C.	905 (x2.87)	T 905
7 recurrence JIT O2	7.04 (x1.39)	37.9 (x1)	T 37.9	C 23.8; N.C.	319 (x1.01)	T 319
8 AmpliCol reference	10.4	101	NOT EXPOSED	19.1	8.21e+03	NOT EXPOSED
8 compiled JIT O1	126 (x12.2)	218 (x2.16)	T 218	30 GB CAP	N/A	N/A
8 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled JIT O3	76.3 (x7.38)	180 (x1.79)	T 180	30 GB CAP	N/A	N/A
8 eager-DAG JIT O2	49.2 (x4.76)	718 (x7.12)	T 716	30 GB CAP	N/A	N/A
8 recurrence JIT O2	66.7 (x6.45)	93.9 (x0.931)	T 93.9	C 1.26e+03; N.C.	6.16e+03 (x0.75)	T 6.16e+03
9 AmpliCol reference	63.4	430	NOT EXPOSED	616	1.05e+05	NOT EXPOSED
9 compiled JIT O1	517 (x8.15)	458 (x1.06)	T 456	30 GB CAP	N/A	N/A
9 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled JIT O3	511 (x8.06)	453 (x1.05)	T 452	30 GB CAP	N/A	N/A
9 eager-DAG JIT O2	113 (x1.78)	531 (x1.23)	T 529	30 GB CAP	N/A	N/A
9 recurrence JIT O2	814 (x12.8)	227 (x0.528)	T 227	1 H CAP	N/A	N/A

Original AmpliCol is the denominator. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value marked n.c.; n.c. means not comparable because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no separately reported evaluator-total timing. Every pyAmpliCol row shows the authenticated accumulated warmed evaluator total marked T; T is not an attribution ratio. Recurrence core/execution attribution remains a separate metric in raw evidence and is not relabeled as evaluator total. Older entries without authenticated total evidence remain marked not exposed; it is not a missing measurement. user cap: native C++/ASM generation is not attempted above n=6 (native-backend-generation-cap-n6-v1).

The $n = 8$ all-flow recurrence retry succeeded after the generic bounded numerical-current evidence fix. Its complete generated artifact was retimed in place after removing a source-label-only runtime rejection; generation took 1263.412s, peak guarded memory was 1.347GB, and all numerical and cross-layout checks passed. The $n = 8$ all-flow JIT O1, JIT O3, and eager-DAG O2 generations are guarded N/A entries: they reached the 30 GB Darwin physical-footprint cap after 2638.855s, 2776.558s, and 2753.840s respectively (peak RSS 6.563 GB, 6.884 GB, and 6.952 GB), before producing artifacts. The $n = 9$ all-flow recurrence generation is a guarded N/A after reaching its 3600-second limit (peak guarded memory 2.320 GB; peak RSS 1.229 GB) before producing an artifact. The $n = 9$ all-flow JIT O1 and JIT O3 generations independently reached the 30 GB cap after 2099.235s and 2049.774s respectively (peak RSS 9.411 GB and 8.988 GB), before producing artifacts. The $n = 9$ all-flow eager-DAG O2 generation likewise reached the 30 GB cap after 2119.015s (peak RSS 8.886 GB), before producing an artifact.